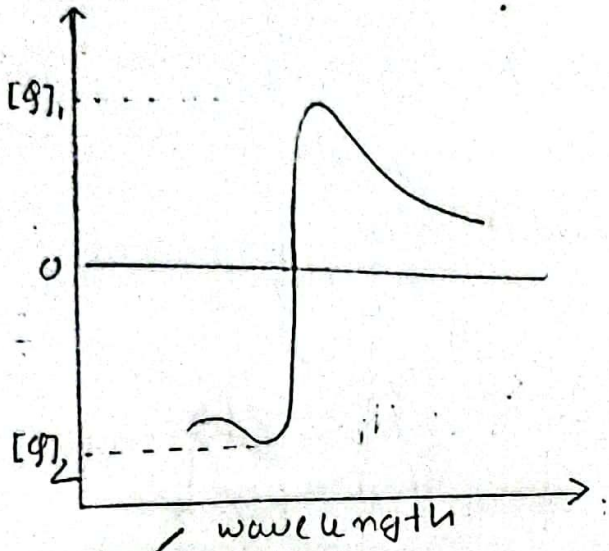
What are the basic differences between the ORD and CD spectroscopy?

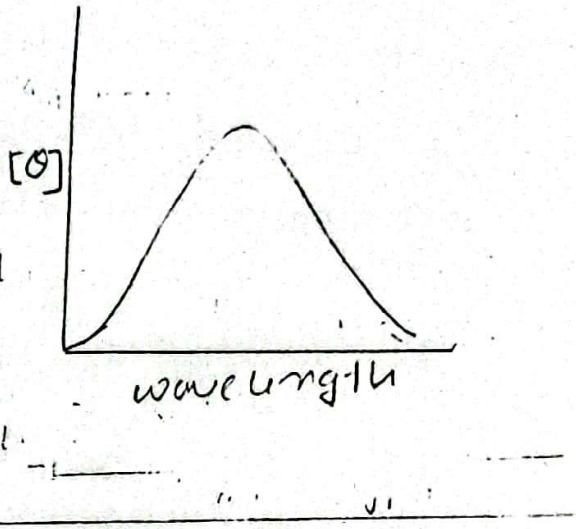
Differences between the ORD and CD spectroscopy:

Optical rotatory dispersion (ORD)	circular dichroism (CD).
1. Measuring optical rotation as a function of wavelength is termed as ORD spectroscopy.	1. CD spectroscopy measures the difference in absorbance of right and left circularly polarized light which arise due to structural assymmetry of a substance.
2. ORD spectra are dispersive.	2. CD spectra are absorptive.
3. In ORD, the circular polarized light is used, is not converted to elliptical light.	3. In CD, the circular polarized light is used and is converted to elliptical light.
4. ORD graps are obtained by plotting specific rotation vs wavelength.	4. CD graps are obtained by plotting molar ellipticity vs wavelength.

5/ Example: ORD curve for camphor.



5. CD curve for camphor.



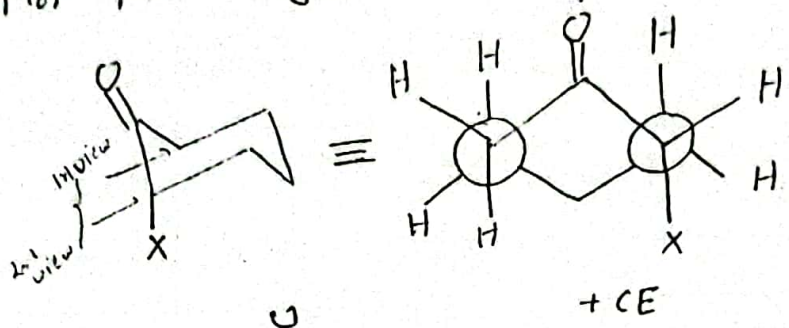
What is axial haloketone rule? Explain the rule and provide its application.

Axial haloketone rule:

"It was suggested that, prediction of the sign of the cotton effect is possible if the ketone group is viewed along the C=O bond in the direction of the ring with the carbonyl carbon at the head of the chair. If the axial α-halogen is found on the right, there exists a positive cotton effect, if it appears on the left, a negative cotton effect is observed."

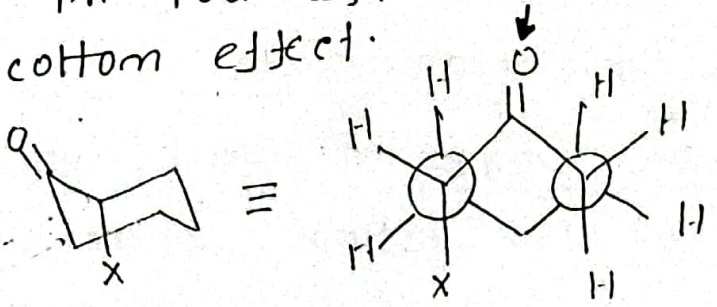
Explanation:

Here, for compound  , we get positive cotton effect. Because when ketone group is viewed along the O=C bond, we get α-halogen in the right side.



+CE

again, for compound  , since we get α-halogen in the left side, there exists negative cotton effect.

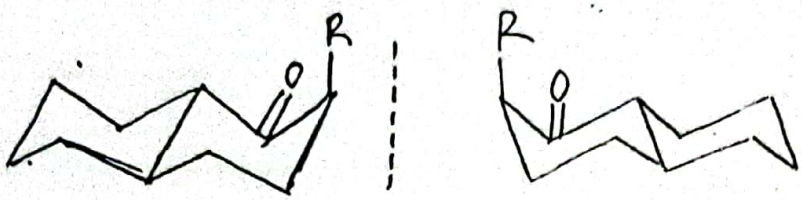


-CE

Application of axial α-haloketone rule:

1. Determination of absolute configuration:

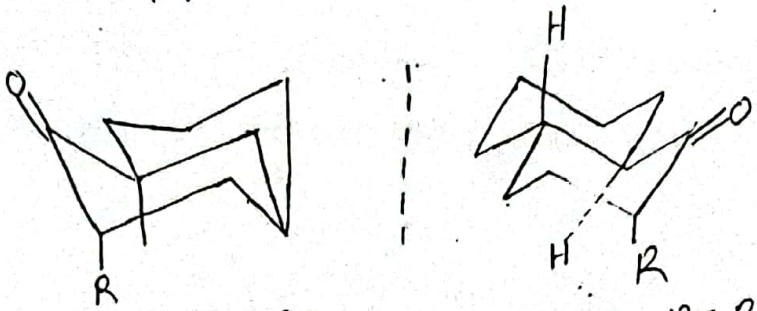
~~Boomo~~ Boomination of (-)-trans-1-decalone gives positive cotton effect which is only possible when the absolute configuration is of (-)-trans-1-decalone is 1a.



1a : R=H

1b : R=H

III



2a : R=Br

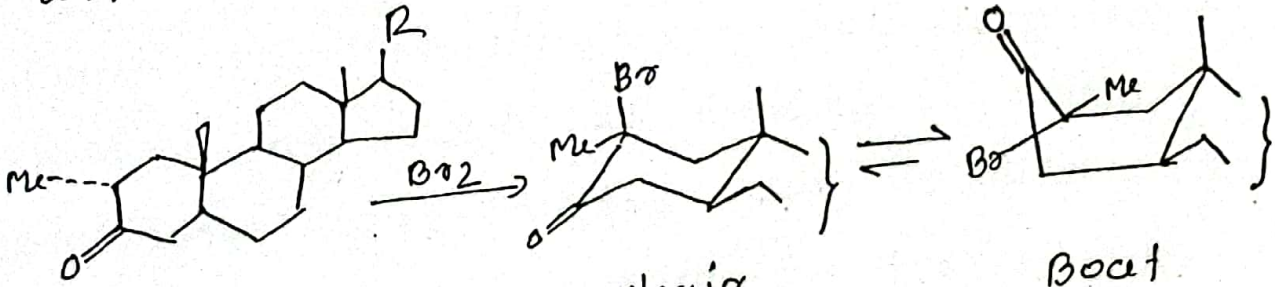
CE = positive

2b : R=Br

CE = negative

2. Determination of boat conformation :

Bromination of 2α-methyl-3-cholestanone gives 2α-bromo-2β-methyl-3-cholestanone with positive cotton effect.



chair

-CE

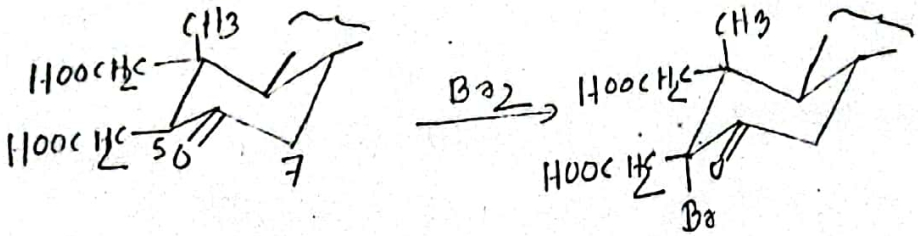
Boat

+CE

This is only possible when the product is in boat conformation.

3. Determination of the position of halogen:

Bromination of the following steroid compound gives 5- or 7- bromo product: ~~with negative~~ when we analysis the product we get negative cotton effect which is only possible when Br is in 5- position.



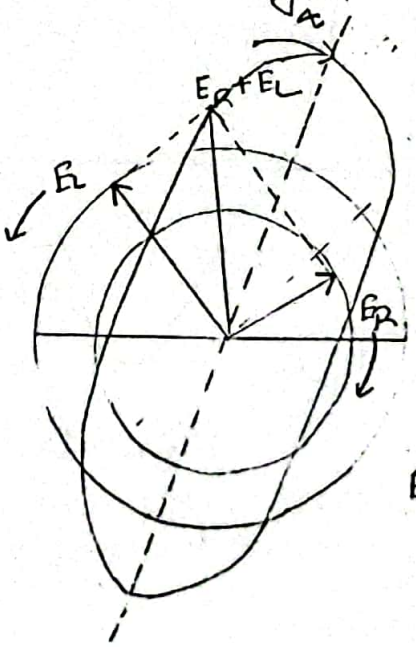
(-) CE

*

Explain the phenomenon circular dichroism.

Circular dichroism:

Circular dichroism is the differential absorption of left and right handed circularly polarized light.



E_R = Right handed circularly polarized light.

E_L = Left handed circularly polarized light.

Right and left-circularly polarized light can be absorbed at different extent at some wavelengths due to differences in extinction co-efficients for the two polarized rays.

The CD curve is dependence of $\epsilon_L - \epsilon_R$ vs λ.

$$\epsilon_L - \epsilon_R = \frac{K_L - K_R}{C}$$

where, ε_L and ε_R are molar absorption co-efficients for left and right handed circularly polarized light and c is the concentration.

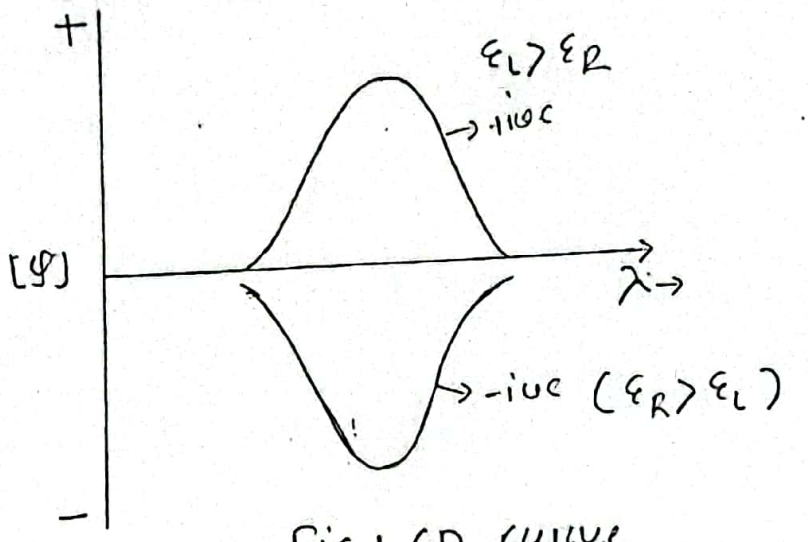


Fig : CD curve

CD curve is based on the following two conditions.

i) when $\varepsilon_L > \varepsilon_R$, CD curve is positive

ii) when $\varepsilon_R > \varepsilon_L$, CD curve is negative.

Draw the two main types of ORD curve.

There are two main types of ORD curve-

(i) Normal or plain curve:

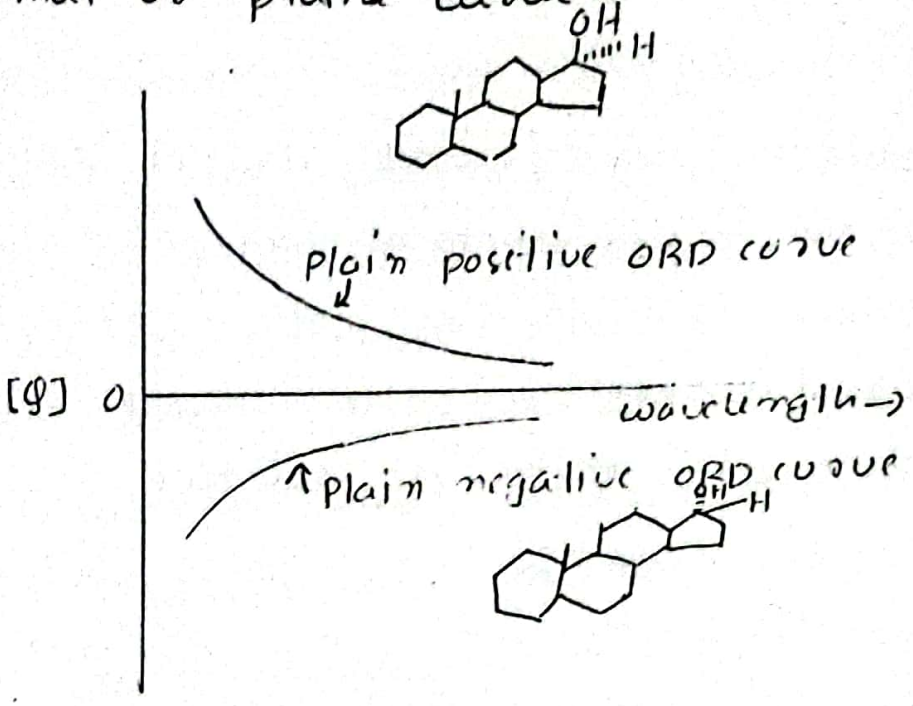
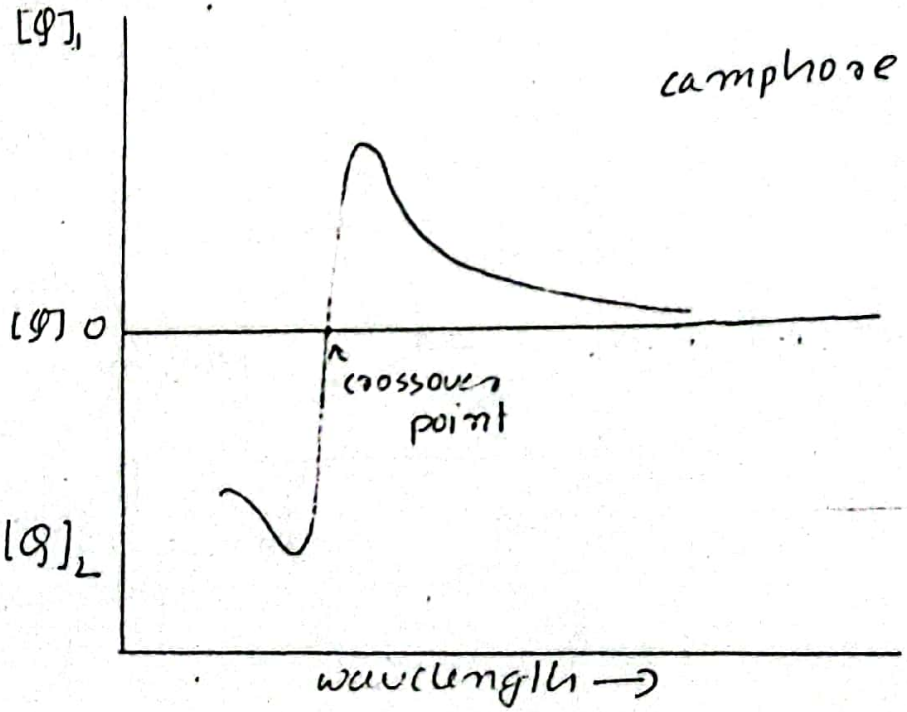


Fig: plain ORD curve.

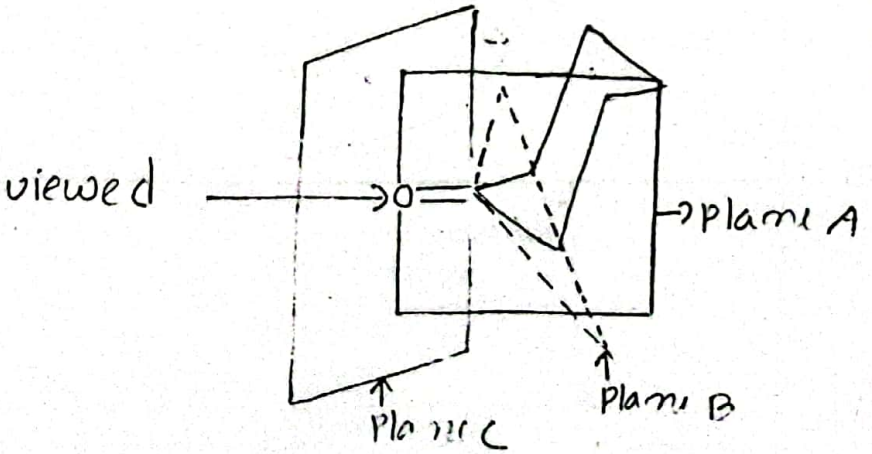
(ii) Anomalous ORD curve:



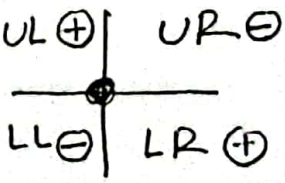
What is octant rule? Explain the rule and provide its application.

Octant rule:

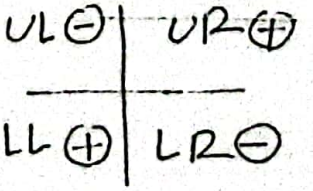
The octant rule stated that, for rear octant, substituents in the lower left and upper right octants make a negative contribution to the cotton effect while substituents in the lower right and upper left make a positive contribution. This is reverse for front octant. substituents in any of the three planes do not make contribution.



For rear octant



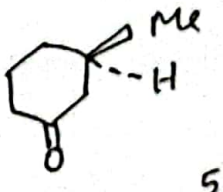
For front octant

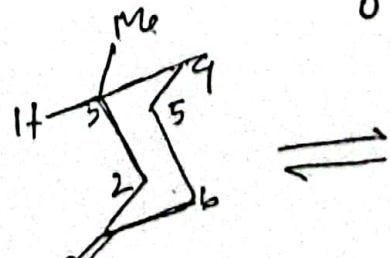


Application of octant rule:

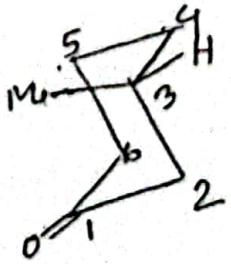
✓ Determination of conformation:

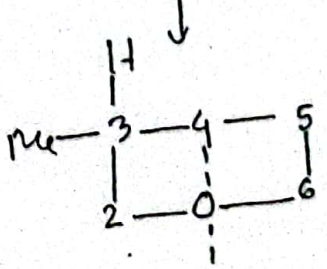
For (+)3-methyl cyclohexanone, we can determine whether the methyl group is in ~~equ~~ equitorial ~~position~~ or axial position. since the the compound has positive cotton effect and by analysis octant rule, we see that positive CE is possible only when methyl group is in equitorial position, so in (+)-3-methyl cyclohexanone, methyl group must be in equitorial position.



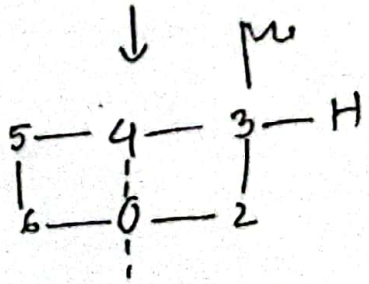


⇌



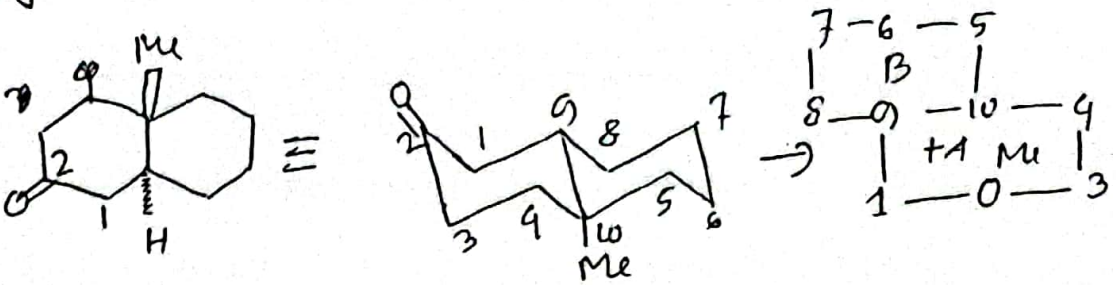


+ve CE



-ve CE.

2. configuration determination:



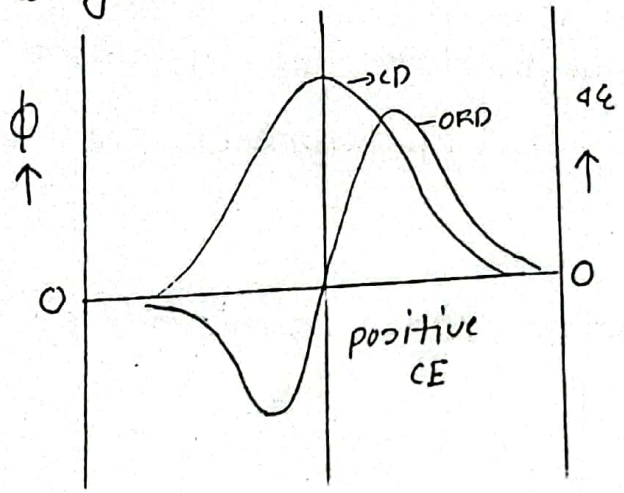
(A)

since for this compound (A), the cotton effect is positive and by analysis of octant rule we get positive cotton effect, so the molecule is in right configuration.

* What is positive cotton effect and negative cotton effect?

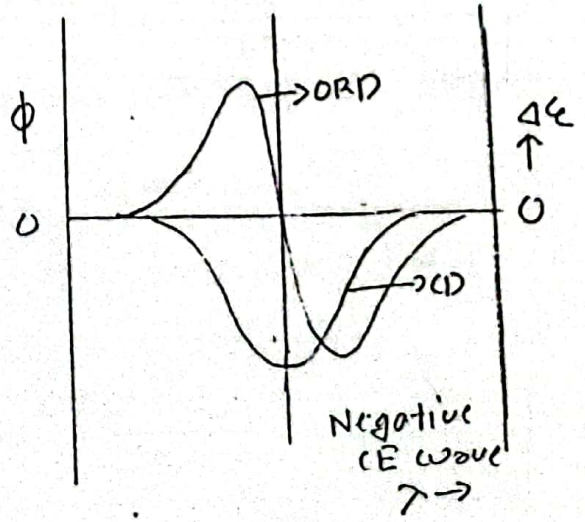
Positive cotton effect:

The cotton effect is called positive if the optical rotation first increases as the wavelength decreases.



Negative cotton effect:

The cotton effect is called negative if the optical rotation first decreases.



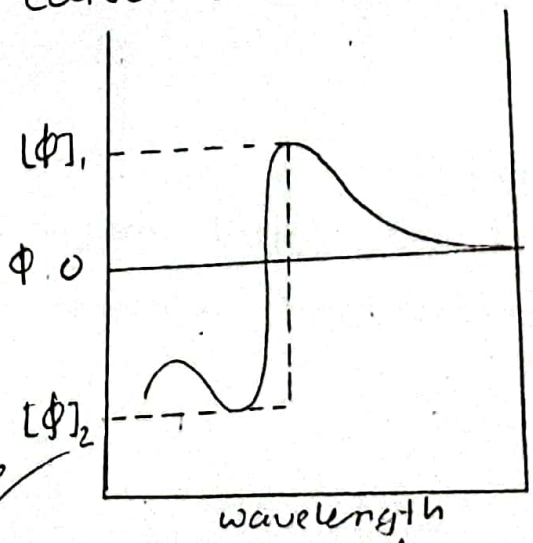
What is cotton effect. Draw CE curve for camphor in both CD and ORD.

cotton effect:

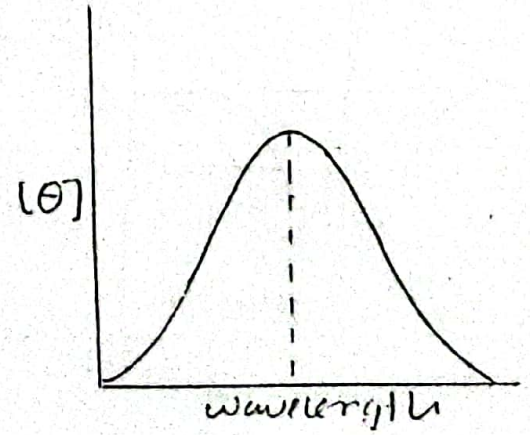
The cotton effect is the characteristic change in optical rotatory dispersion and/or circular dichroism in the vicinity of an absorption band of a substance.

The combination of circular dichroism and circular birefringence is known as cotton effect.

ORD curve for camphor:



CD curve for camphor:



* phenomenon of optical rotatory dispersion:

The variation of optical rotation as a function of wavelength is called optical rotatory dispersion.

If linear polarized light passed through a sample for which the refractive indices of the left and right handed polarized components differs, the components will upon recombination give plane polarized radiation in which the plane of polarization is rotated by an angle α, given by -

$$\alpha = \frac{n_L - n_R}{\lambda}$$